

Lemma 3.2. *Let M be an R -module. Suppose that I is an ideal of R and $N \in \text{mod } R$ such that $\text{Supp } N \subset V(I)$. Then the following assertions hold for any non-negative integer n .*

- (1) *If $\text{Ext}_R^i(R/I, M) \in \mathcal{S}$ for $0 \leq i \leq n$, then $\text{Ext}_R^i(N, M) \in \mathcal{S}$ for $0 \leq i \leq n$.*
- (2) *If $\text{Tor}_i^R(R/I, M) \in \mathcal{S}$ for $0 \leq i \leq n$, then $\text{Tor}_i^R(N, M) \in \mathcal{S}$ for $0 \leq i \leq n$.*

Proof. Since N is finitely generated, it follows from [13, Theorem 6.4] that there exists a chain $0 = N_0 \subset N_1 \subset \cdots \subset N_k = N$ of submodules of N such that for each i we have $N_j/N_{j-1} \cong R/\mathfrak{p}_j$ with $\mathfrak{p}_j \in \text{Supp } N$.

(1) Note that $\text{Hom}_R(R/I, E)$ is an injective R/I -module for any injective R -module E by [12, Lemma 3.5]. For each $\mathfrak{p}_j$, by [15, Theorem 10.64], there is a spectral sequence

$$E_2^{p,q} = \text{Ext}_{R/I}^p(R/\mathfrak{p}_j, \text{Ext}_R^q(R/I, M)) \Rightarrow \text{Ext}_R^n(R/\mathfrak{p}_j, M).$$

Being a subquotient of a finite direct sum of copies of $\text{Ext}_R^q(R/I, M) \in \mathcal{S}$, we have that $E_2^{p,q} \in \mathcal{S}$ for $0 \leq q \leq n$ and any p . Since $E_\infty^{p,q}$ is isomorphic to a subquotient of $E_2^{p,q}$, we have $E_\infty^{p,q} \in \mathcal{S}$ for $0 \leq q \leq n$ and any p . Hence $\text{Ext}_R^i(R/\mathfrak{p}_j, M) \in \mathcal{S}$ for $0 \leq i \leq n$. Consequently, applying the functor $\text{Ext}_R^i(-, M)$ to exact sequences $0 \rightarrow N_{j-1} \rightarrow N_j \rightarrow N_j/N_{j-1} \rightarrow 0$ yields that $\text{Ext}_R^i(N, M) \in \mathcal{S}$ for $0 \leq i \leq n$.

(2) can be proved similarly by using [15, Theorem 10.60]. $\square$

(3.3) Let $I = (x_1, x_2, \dots, x_n)$ be an ideal of R and M a complex of R -modules. In the following, we use $K(I)$ to denote the Koszul complex with respect to $I = (x_1, x_2, \dots, x_n)$. The Koszul complex $K(I)$ is a bounded complex of finite free modules. Then we define $H_i(K(I), M) := \text{Tor}_i^R(K(I), M) = H_i(K(I) \otimes_R M)$ and $H^i(K(I), M) := \text{Ext}_R^i(K(I), M) = H^i(\text{Hom}_R(K(I), M))$.

(3.4) The preceding lemma allows us to extend some parts of [3, Theorem 2.8] to modules which are not necessarily finitely generated.

Proposition 3.5. *Let M be an R -module. If I and J are ideals of R such that $\sqrt{I} = \sqrt{J}$, then the following assertions are equivalent for any non-negative integer n .*

- (1) $\text{Ext}_R^i(R/I, M) \in \mathcal{S}$ for $0 \leq i \leq n$.
- (2) $\text{Ext}_R^i(R/J, M) \in \mathcal{S}$ for $0 \leq i \leq n$.
- (3) $H^i(K(I), M) \in \mathcal{S}$ for $0 \leq i \leq n$.
- (4) $H^i(K(J), M) \in \mathcal{S}$ for $0 \leq i \leq n$.

Proof. (1) $\Leftrightarrow$ (2) follows from Lemma 3.2(1).

(1) $\Rightarrow$ (3) There is a spectral sequence

$$E_2^{p,q} = \text{Ext}_R^q(H_p(K(I)), M) \Rightarrow H^{p+q}(K(I), M).$$